

To: Divyanshu

Subject: Macro regime system v2 — refinements, transition states, HMM layer, and combo discovery methodology

From: Rohit

This is the consolidated plan for the macro regime model following our review. It supersedes earlier piecemeal notes. I have organised it as defined, achievable steps in priority order. Read all of it and ask me your questions...

The goal is a regime system that is clean, dynamic, and highly likely to backtest well — consistent with the MindWealth demo and the macro agent demo output.

PART A — REFINING THE 5 REGIME DIMENSIONS

The five dimensions (rate environment, credit conditions, geopolitical backdrop, valuation, monetary accommodation) are a sound proprietary synthesis — roughly the Bridgewater All-Weather logic (growth, inflation, policy) with a geopolitical overlay. Keep the five. Refine each as follows.

A1. fed_cycle — reduce from 7 states to 4

The current set (HIKING_EARLY/LATE, CUTTING_EARLY/LATE, PAUSING, QE, QT) is too granular for our sample sizes. Collapse to: TIGHTENING (hiking or holding high), PIVOTING (first cut just occurred), EASING (cuts in progress), EASY (at/near zero). This roughly doubles observations per state. Keep QE/QT as a separate liquidity flag (see A5), not a fed_cycle state — balance-sheet policy and rate policy are different levers.

A2. curve_regime — keep 4 states, add fiscal caveat

Keep INVERTED, FLAT, STEEPENING, NORMAL. The 2022–23 episode (deepest inversion in 40 years, no recession) challenges the pure inversion signal. The curve works through the credit transmission mechanism, which was offset in 2022–24 by AI capex and large fiscal deficits. Add a conditioning note to the classifier: the curve recession signal is materially weaker when the fiscal deficit exceeds 5% of GDP.

A3. val_regime — keep 4 states, add rate-of-change

Keep EXTREME, ELEVATED, FAIR, CHEAP. CAPE has been ELEVATED or EXTREME almost continuously since 2017, so a static level gate loses discriminatory power. A fresh crossing into EXTREME is a different signal from sitting at EXTREME for three years. Store both the level tier and the 6-month change in percentile rank.

A4. geo_overlay — simplify from 6 states to 3

Claude API classifying fine-grained geopolitical states from a date is noisy (SANCTIONS vs TRADE_WAR is blurry). Collapse to: NEUTRAL, ELEVATED_RISK (trade wars, sanctions, regional conflicts), CRISIS (pandemic, major financial crisis, global-war-level events). Three states classify far more reproducibly.

A5. liquidity — expand from 2 states to 4

GLOBAL_EASY/GLOBAL_TIGHT is too binary; the direction of change matters as much as the level. Use a 2×2: easy-and-improving, easy-and-tightening, tight-and-improving, tight-and-tightening. WALCL direction distinguishes these. A Fed hiking while running QT is tighter than a Fed hiking with a flat balance sheet.

DELIVERABLE A: Updated classifier prompt (Section 5.2) with the refined state lists, the fiscal caveat for curve_regime, and CAPE rate-of-change added to val_regime. Re-run the historical backfill and report the distribution of labels across history so we can check no state is degenerate.

PART B — THE 14TH VARIABLE AND HISTORY WINDOWS

B1. New variable — 2-year US Treasury yield 8-week rate of change (TWY_ROC, variable #14)

Role: Regime classification ONLY. Not a new combo-firing signal. Do NOT include TWY_ROC in the 13Cn combo enumeration — it informs regime classification and the conviction modifier only.

Full variable spec:

- **Variable ID:** TWY_ROC (variable #14)
- **Name:** 2-Year Treasury Yield, 8-week rate of change
- **Source:** FRED DGS2 (2-year constant-maturity Treasury yield, daily). Use the Friday close to align with our weekly cadence.
- **Paradigm:** ROC (rate-of-change), consistent with WTI and CNH. Not a level, not a dual.
- **Calculation:** $TWY_ROC = DGS2(\text{today}) - DGS2(56 \text{ calendar days ago})$, expressed in percentage points (pp). Positive = yields rising = market pricing a more hawkish path; negative = yields falling = market pricing cuts/easing.
- **Direction bands for classifier prompt:** HAWKISH if $TWY_ROC > +0.30pp$; NEUTRAL if $-0.30pp$ to $+0.30pp$; DOVISH if $< -0.30pp$. Backtest these bands — $\pm 0.30pp$ is a starting point, not validated.
- **History window:** 3-year rolling (it is a rate-of-change/flow variable, same group as WTI, CNH, WALCL, CPI).
- **Inflation adjustment:** None — already a change in a rate, scale-invariant.
- **Why it matters:** Every other fed-cycle input (FFR, WALCL, curve, NFCI, CAPE) is coincident or lagging. TWY_ROC LEADS Fed action by roughly 3–9 months because bond markets price in dot plots, speeches, and forward guidance automatically. When fed_cycle and TWY_ROC diverge (classifier says TIGHTENING but TWY_ROC is strongly DOVISH), that divergence is itself the most actionable signal — the market is pricing a pivot before the lagging variables show it.
- **Feed into classifier prompt as:** "2yr yield 8wk change: {TWY_ROC}pp, direction: {HAWKISH/NEUTRAL/DOVISH}."
- **Validation to run when you build it:** On 7 April 2025 (the tariff-shock market bottom), DGS2 had fallen roughly 65–75bps over the prior 8 weeks — strongly DOVISH — while the fed_cycle label would still have read TIGHTENING/PAUSING. The leading variable called the bottom; the lagging label did not. Confirm this against the actual DGS2 series.

B2. History windows — confirm and implement

This supersedes the earlier 3-year-rolling-for-everything spec:

- **Structural/level variables** (CAPE, VIX, yield curve, NFCI, GSR): FULL EXPANDING HISTORY from inception. Never rolling. VIX from 1990, CAPE from 1881, curve back far enough to capture 1970s–80s inversions.
- **Flow/rate-of-change variables** (WTI 4wk%, CNH 4wk%, WALCL MoM%, CPI surprise, and the **new TWY_ROC**): 3-year ROLLING.
- **Store BOTH** `unconditional_pctile` (full history) and `regime_pctile` (conditioned on `fed_cycle`) for every variable every day. Combo detection uses unconditional. The **conviction modifier** uses `regime_pctile`. Fall back to unconditional if the regime-conditioned subset has fewer than 50 observations, and log which was used.

B3. Triple-storage for velocity-sensitive variables

For variables where both static level AND velocity matter (CAPE is the clearest case), store three numbers: full-history expanding rank, 3-year rolling rank, and 8-week rate-of-change of the rank. **We then test which combination predicts forward returns best rather than committing a priori.**

DELIVERABLE B: Variable table updated to 14 variables with correct window per variable, TWY_ROC fully specced, dual-storage (unconditional + regime) implemented, and triple-storage (level/rolling/velocity) for CAPE.

PART C — PERCENTILE RANK AS EMISSION PROBABILITY

For each variable each day, compute one number: where today's reading sits in that variable's own historical distribution (VIX at 35 against 1990-present is roughly the 92nd percentile = 0.92). **The 14 percentile ranks form the daily observation vector.**

Today, a variable below its RARE threshold contributes nothing — it is binary, fire or not. The emission-probability approach treats every reading as partial evidence about which regime we are in. A VIX at **0.72** (below the **0.80 RARE threshold**) still provides moderate evidence for a stressed regime. Sub-threshold readings accumulate week by week, so the model becomes sensitive to a regime building before any single variable crosses its line.

This is Bayesian updating made explicit: $P(\text{regime} \mid \text{today's readings}) = P(\text{today's readings} \mid \text{regime}) \times P(\text{regime}) / P(\text{today's readings})$. **$P(\text{regime})$ is yesterday's posterior (the prior).** **$P(\text{today's readings} \mid \text{regime})$ is the emission probability.** **The output is the new posterior.** We act proportionally to the posterior distribution across regimes — never snapping to a single label.

DELIVERABLE C: Store the 14 percentile-rank vectors daily alongside `combo_fires`. No HMM yet — you need 6+ months of clean stored vectors before HMM training is meaningful. This storage is the foundation for Part D.

PART D — THE HMM LAYER (build AFTER C has accumulated data)

D1. Layer, do not replace

Do NOT replace the percentile engine. Layer a simple 3-state HMM on top: Risk-On, Risk-Off, Transition. Observations = the 14 daily percentile-rank vectors. Output = a daily posterior, e.g. [Risk-On 0.65, Risk-Off 0.25, Transition 0.10].

D2. Feed posteriors into the classifier prompt

Add to the Section 5.2 prompt: "HMM posterior — Risk-On 0.65, Risk-Off 0.25, Transition 0.10. Use as a soft prior. If Risk-Off probability exceeds 0.40, weight toward tighter classifications even if individual variables have not crossed thresholds." This gives the classifier a model-synthesised view of whether the overall pattern matches historical stressed periods.

D3. Detecting distribution shifts

The HMM compares whether today's observations are being generated by the same distributions it learned in training. If HY OAS sits at the 65th percentile for 8 straight weeks in a period where the Risk-On state treats 65th-percentile HY as unlikely, the model accumulates evidence against Risk-On each week. Once the posterior for the assumed regime drops below 50%, you have detected a regime shift — earlier than waiting for any single variable to breach its 85th-percentile threshold. How different is different enough is answered by the posterior crossing 50%, not a hand-set cutoff.

D4. Regime duration — Hidden Semi-Markov extension

A plain HMM is memoryless about how long you have been in a state. An HSMM adds an explicit dwell-time distribution per state, mapping directly to our combo durations (Combo C SHORT <6wks / MEDIUM 6–16wks / LONG >16wks). The HSMM answers "is this regime ending?" as $P(\text{exit within } N \text{ weeks} \mid \text{already in state } M \text{ weeks})$ — exit probability rises steeply as M passes the typical dwell time. Treat as a phase-2 enhancement once the basic HMM is validated.

DELIVERABLE D: 3-state HMM trained on accumulated vectors; daily posterior stored and piped into the classifier prompt; a backtest showing whether regime-conditioning the signals via the posterior improves Sharpe, win rate, and drawdown versus no regime overlay. Start from the existing regime_backtest.py harness.

PART E — TRANSITION PROBABILITY AS AN OPTIONS PROBLEM

This SUPPLEMENTS all qualitative "the combo may be clearing" talk with real probabilities. Treat each combo CANCEL condition as a digital barrier option.

For Combo C: cancel = "WTI 4-week change < +5% for 4 consecutive Fridays AND CPI not hot for 2 consecutive prints."

WTI leg: for each upcoming Friday, the strike is the WTI price 4 weeks before that Friday $\times 1.05$. Cancel for that Friday is met if WTI closes below the strike. Per-Friday probability = $N(-d_2)$, where $d_2 = [\ln(S/K) + (-\sigma^2/2)T] / (\sigma\sqrt{T})$, S = current WTI, K = strike, σ = annualised WTI vol (~35%), T = weeks to that Friday. Because the four 4-week windows overlap (sharing 3 of 4 weeks, correlation ~0.75), do NOT multiply the four marginals. Run a Monte Carlo with GBM paths to capture the correlation exactly.

CPI leg: a separate discrete binary — historically what fraction of CPI prints come in at-or-below consensus? That per-print probability squared approximates $P(2 \text{ consecutive non-hot prints})$. Multiply the WTI-leg probability by the CPI-leg probability to get total $P(\text{Combo C fully cancels})$.

Generalises to any rule: S = current variable, K = threshold, σ = its vol, T = horizon, correlation = window overlap. Combo F's 26-week window is deterministic. Combo D's 3–10 day horizon is a fast-expiring digital. **Combo G's "leads vol spike 3–6 weeks" is effectively a variance option.**

DELIVERABLE E: A reusable function `combo_cancel_probability(variable, threshold, current_value, vol, weeks_remaining, consecutive_required)` returning the Monte Carlo probability. Wire to display a live cancel/persist probability for every active combo on the nightly briefing and dashboard.

PART F — FORMALISING REGIME DEFINITIONS THAT ARE CURRENTLY VAGUE

F1. TIGHTENING-LATE (formerly HIKING_LATE)

Must have explicit quantitative triggers. Proposed: FFR > 3.5% AND FFR risen > 150bps in prior 12 months AND ($T_{10Y2Y} < -30\text{bps}$ OR hike pace decelerating over last two meetings). October 2022 confirms this (FFR ~3.25%, +300bps in 9 months, curve -50bps). Add explicit rules for every `fed_cycle` state and validate against the backfill.

F2. INVERTED — definition (not yet in any document; define here)

Currently only a Claude-API-classified label with no numeric rule, and the steepening logic references "post-inversion" without defining inversion. This single standard must be used everywhere — by both this workstream and Ahil's steepening-of-inversion short gate, so the two are consistent. Source variable: T_{10Y2Y} (FRED).

- **Inversion event:** T_{10Y2Y} crosses below 0bps.
- **INVERTED regime:** $T_{10Y2Y} < 0\text{bps}$ sustained for ≥ 4 consecutive weeks. A one-day dip below zero is not an inverted regime.
- **Trough:** the most negative T_{10Y2Y} value reached during the inverted regime.
- **Relationship to variable #9 thresholds:** -30bps is the RARE tier and -80bps the EXTREME tier for the curve variable (combo detection) — these are separate from the regime label. The regime label INVERTED uses the <0bps/4-week rule above; the variable thresholds remain as specced for combo firing.

F2a. STEEPENING regime — defined consistently

After a trough in an inverted regime, T_{10Y2Y} rises $\geq +15\text{bps}$ over 4 weeks (RARE, per variable #9) or $\geq +40\text{bps}$ over 4 weeks (EXTREME). The `curve_regime` classifier should apply these numeric rules rather than inferring STEEPENING from context, so the label is reproducible and testable against the backfill.

F3. HIKING period (for the beta/directionality test)

FFR rising AND cumulative hike cycle > 100bps.

F4. Steepening-of-inversion short trigger

Now inherits the precise inversion definition from F2. Valid when: (a) the prior inverted regime had a trough deeper than -50bps; (b) the curve is now steepening ($\geq +15\text{bps}/4\text{wk}$); (c) no large **fiscal/QE** offset is active (fiscal deficit < 5% GDP AND no active QE). It failed in 2022–23 precisely because the -108bps inversion was offset **by AI capex and fiscal spend**. Backtest

trough threshold (−50 vs −80bps) and steepening threshold (+15 vs +40bps/4wk) before committing.

Important note on evidence standard: This gate rests on a documented causal mechanism plus the 2000/2007 analogs and the understood 2022–23 failure — NOT on a win-rate statistic. It must be justified by mechanism and analog consistency, never by "it backtested well over 2–3 fires." If the only argument for it becomes the backtest number, it falls back under the ≥ 5 -instance statistical rule. This distinction is the same one we apply to deep-inversion in Ahil's short-gate work: at genuine tail extremes, small consistent samples plus causal logic are the appropriate evidence standard; large samples are reserved for the core repeatable rules.

DELIVERABLE F: Explicit quantitative definitions for all fed_cycle states, the INVERTED regime (F2), the STEEPENING regime (F2a), the HIKING period (F3), and the steepening-short trigger (F4) — all validated against the backfill, and all using T10Y2Y consistently with Ahil's gate.

PART G — PERSISTENCE SIGNALS (define and wire — currently only named in the Addendum)

G1. SEVEN_WEEK_GRIND

SPX weekly close higher than prior week by at least +0.5% for 7 consecutive weeks. **Saturday job.** Wire as an **AMPLIFIER** to Combo E (valuation extreme) — when both fire, the structural warning is elevated. It is NOT a standalone short signal: across history the average 6-month return after a grind fires is positive and the great majority of instances continued higher.

G2. VIX_SUPPRESSED

VIX close < 15 for 10 consecutive trading days. Daily job. Wire as a **PRECURSOR/WATCH FLAG** to Combo D (FOMO/euphoria). It precedes vol events only about half the time historically, so it is a watch-list flag, not a sell trigger.

For both, store start_date, most_recent_date, streak_length, and combo_link in a persistence_fires table, and surface on the nightly briefing as "[SIGNAL] active for N periods — monitor for Combo D/E confirmation."

DELIVERABLE G: persistence_fires table populated by the two jobs, wired to the briefing with the correct amplifier/precursor framing.

PART H — COMBO DISCOVERY METHODOLOGY (the full pipeline)

Build this as an explicit, automated pipeline so it runs without hand-holding.

1. **Detection.** For each of the 298 combinations (13C1 + 13C2 + 13C3 — note this stays 13, NOT 14; TWY_ROC is a regime input, not a combo variable), compute the full-expanding-history percentile rank of each required variable. Record a fire when ALL required variables are simultaneously at-or-beyond their RARE threshold. Store date, variable IDs, combo type, and named-combo label (A–G or null).

2. **Forward returns.** For each historical fire date, store SPX forward return at 1m/3m/6m/9m/12m. The classic research-desk table: combo fires, and what happened next over each horizon.
3. **Regime tagging.** For each fire date, attach the 5 regime labels (from the backfill) so hit rate can be sliced by regime.
4. **Surfacing gate.** A combo surfaces as a candidate at ≥ 3 fires AND $\geq 60\%$ hit rate at the relevant horizon. Hit-rate definition: bullish combo = fraction of fires where SPX forward return > 0 ; bearish combo = fraction where forward return < 0 . At genuine extremes we deliberately accept small samples — the premise is that at tail extremes you do not need large N. Large N is reserved for the core asset-function rules.
5. **Beta / luck filter.** A combo that fires mostly in easy-money bull periods may just be capturing market drift. Test: regime-conditional hit rate in HIKING or INVERTED periods must still be $\geq 55\%$. Why 55% and not 60–65%: HIKING/INVERTED periods are structurally adverse for longs, so demanding 60%+ there would wrongly kill genuine signals that merely perform a little worse in hostile regimes. Report 55% and 60% side by side and decide per combo. IMPORTANT CAVEAT: a 2-of-3-legs version underperforming the 3-of-3 version is NOT by itself proof the combo is "just beta" — the third leg may genuinely add signal. Treat 2-of-3 vs 3-of-3 as one diagnostic among several, not a verdict. Compare each combo's forward returns against (a) the unconditional base rate, (b) days with similar single-variable extremes but without the full combo, and (c) the regime-conditional base rate. Genuine signal must beat all three.
6. **Directionality consistency.** Require hit rate $\geq 50\%$ across at least 2 of the 5 regime-dimension categories. A combo with 100% hit rate in QE and 20% in HIKING is a regime-beta artifact, not a signal.
7. **Economic story (Tavila).** For each surviving candidate, the Tavila agent retrieves what was actually happening on each historical fire date (e.g. "Draghi whatever-it-takes, Italy 2011"). You eyeball whether the combo has a coherent causal story or is coincidence. Strong stats with no economic narrative = treated with suspicion.
8. **Naming gate (promotion).** A candidate is promoted to a named/validated combo only at ≥ 5 fires AND $\geq 80\%$ hit rate AND passing the beta filter AND a coherent story. (This matches the Named Combos doc standard: $\geq 80\%$ hit rate / ≥ 5 instances. The 60%/3-fires surfacing tier in step 4 is the softer first pass; 80%/5-fires is the promotion bar.)
9. **Output table.** For each named combo: required variables and thresholds, direction, horizon, hit rate, average return per horizon, historical instances with their stories, cancel condition, and current cancel probability (from Part E).

DELIVERABLE H: The full 9-step pipeline automated, candidate review running monthly, beta/story filters built in as steps 5–7. Claude API does step 7 and assists steps 5–6 reasoning; the arithmetic of steps 1–4 is pure Python/SQL.

PART I — SAMPLE-SIZE DISCIPLINE AND HONEST FRAMING

- I1. Never make an investment decision on a regime-conditional percentile computed from fewer than 30 observations. Below 50, fall back to unconditional and log it. The 3–4 instances of a combo firing in a specific regime since 1990 is too thin for statistics alone — which is why steps 5–7 (base-rate comparison + economic story) carry the weight at extremes, not the raw hit rate.
- I2. Two evidence standards, used deliberately for two different types of claims:

- **Statistical gate (SSI/variable gates, unnamed combos):** ≥ 5 instances to trust the win-rate statistic. The evidence IS the statistic, so you need enough independent fires to trust it.
- **Mechanism + analog gate (deep-inversion, named combos at extremes):** fewer instances are acceptable only because each one is a legible causal episode, not an anonymous data point. The evidence is causal soundness + consistency of the analogs, not the win-rate number. If you ever catch yourself defending such a gate solely by its backtest win rate rather than its mechanism, it has quietly switched into the statistical bucket and the ≥ 5 -instance rule should bite.

I3. Compound-probability framing for external presentation: requiring several variables to be simultaneously extreme, validated against the 4–10 historical instances where that AND-condition performed at 80–85%, IS a form of compound/joint probability applied deliberately to small samples at extremes. That is defensible because it is at extremes and because we demand an economic story. State it that way: at genuine tail extremes, small consistent samples plus causal logic are the appropriate evidence; large samples are reserved for the core repeatable rules.

SEQUENCING

10. Part B (14th variable + windows) and Part C (store percentile vectors) — foundational, do first.
11. Part F (formal definitions) and Part G (persistence signals) — quick wins, no data accumulation needed.
12. Part E (options-style cancel probability) — independent of HMM, do early.
13. Part A (refine 5 dimensions) — re-runs the backfill.
14. Part H (combo pipeline with beta/story filters) — the core methodology.
15. Part D (HMM layer) — last, needs 6+ months of stored vectors from Part C.

Do not over-engineer — every item above earns its place by being likely to improve backtested performance, not by looking sophisticated.

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